

6.1 RATIO

The word ratio is commonly used in our daily life while distributing the quantities and money and also in household activities. Comparison of two quantities of the same kind is called ratio. This comparison is always in the form of numbers. A ratio is represented by a fraction or the symbol “ : ”. So, if the cost of two toffees is Rs 2 and Rs 3 then their ratio will be $\frac{2}{3}$ or 2:3, read as the “ratio of 2 is to 3”. Similarly $a:b$ read as the “ratio of a is to b ”. We can also express ratio for more than two quantities as 3:5:4 or $a:b:c$.

Define ratio as a relation which one quantity bears to another quantity of the same kind with regard to their magnitudes

Let us learn more about ratio from the following Activity.



Activity

Weights of Sahar and Shabana are 45 kg and 15 kg respectively. To relate their weights, consider the following statements.

- Sahar is 30 kg heavier than Shabana.
- Sahar's weight is 3 times of Shabana's weight.
- Shabana is 30 kg lighter than Sahar.
- Shabana's weight is $\frac{1}{3}$ of Sahar's weight.

Now answer the following:

- Which of the above statements are showing comparison by addition and subtraction?
- Which statements are showing comparison by multiplication and division?

Teacher's Note

Teacher should help the students to enhance the concept of ratio by giving more examples from daily life.

In the above activity the ratio of magnitudes of Sahar to Shabana's weight is 45:15, where as of Shabana to Sahar's weight is 15:45 which are different ratios.

The order in which a ratio is written is important.

How can we find the ratio between 3 km and 21 m?

We use ratio to compare quantities of the same kind and same units. Such as magnitudes of two lengths, two masses, two volumes etc. Consider the following.

The ratio between quantities of 3 bats and two dolls cannot be written as 3:2 because bat and doll are not of the same kind but we can establish a ratio between 3 bats and 2 bats as 3:2 and 2 dolls and 3 dolls as 2:3

Know that a ratio has no units

When ratio between two quantities is established, no unit is written with the ratio, as ratio represents a number.

For Example:

If Arif has 9 rupees and Saleem has 20 rupees then ratio of their amounts is written as 9:20 (with no unit mentioned)

Example 1.

The age of Amaan is 9 months. The age of his elder sister is 3 years. The relation between their ages cannot be written as 9:3 because the age of Amaan is given in months and the age of his sister is given in years, these are two different quantities.

So, age of his sister must be converted into months.

As 3 years = 36 months

So, the ratio between the ages of Aman to his sister will be 9:36

To find ratio between two quantities it is necessary that they must be of the same kind.

Hence we can conclude that, ratio is a relation in which one quantity bears to another quantity of the same kind with regard to their magnitudes.

Know that of the two quantities forming a ratio, the first one is called antecedent and the second one consequent

If a and b are two quantities of the same kind, the ratio between them will be written as $a:b$. Here the first element a is called the antecedent and the second element b is called consequent.

For Example:

Ratio	Antecedent	Consequent
2:3	2	3
5:7	5	7
18:23	18	23

Calculate ratio of two numbers

Example: Find the ratio of the following:

- (i) Ali has Rs 500 and Seema has Rs 309. Write the ratio of their amounts.

Solution: The required ratio is 500:309

- (ii) Nazir get 35 litres of fuel filled in his car and Aijaz filled 48 litres of fuel in his car. Write the ratio of quantities of their fuel.

Solution: The required ratio is 35:48

Reduce given ratio into lowest (equivalent) form

We can find lowest equivalent ratios by dividing each element of the ratio by the same number.

Example 1: Find equivalent ratios of 40:8

Solution:

$$\begin{aligned} & 40 : 8 \\ &= 40 \div 2 : 8 \div 2 \quad (\text{Dividing by 2}) \\ &= 20 : 4 \end{aligned}$$

$$\begin{aligned}\text{or } 40 : 8 \\ &= 40 \div 4 : 8 \div 4 \quad (\text{Dividing by 4}) \\ &= 10 : 2\end{aligned}$$

$$\begin{aligned}\text{or } 40 : 8 \\ &= 40 \div 8 : 8 \div 8 \quad (\text{Dividing by 8}) \\ &= 5 : 1\end{aligned}$$

So, $20 : 4$, $10 : 2$ and $5 : 1$ are some of the lower equivalent ratios of $40 : 8$.

Note: In the above example, $5 : 1$ is the **lowest equivalent form** of $40 : 8$ as both 5 and 1 are natural numbers having no common factor.

Example 2: Find the ratio between Rs 210 and Rs 105 and write in lowest equivalent form.

$$\begin{aligned}\text{Solution: Ratio} &= 210 : 105 \\ &= 70 : 35 \quad (\text{Dividing both by 3}) \\ &= 14 : 7 \quad (\text{Dividing both by 5}) \\ &= 2 : 1 \quad (\text{Dividing both by 7})\end{aligned}$$

Which is the required ratio.

Example 3: Find the ratios of the following quantities and write in lowest form.

- (i) 15 kg and 700 g (ii) 9 hours and 36 minutes.

Solution:

- (i) In the given ratio 15 kg : 700g, the units are different.
As 1 kg = 1000 g

$$\text{Therefore, } 15 \text{ kg} = (15 \times 1000) = 15000 \text{ g}$$

Hence, the ratio in the lowest form

$$\begin{aligned}&= 15 \text{ kg} : 700 \text{ g} \\ &= 15000 \text{ g} : 700 \text{ g} \\ &= 15000 : 700 \\ &= 150 : 7 \quad (\text{dividing both by 100})\end{aligned}$$

(ii) 9 hours and 36 minutes

As 1 hour = 60 minutes

Therefore, 9 hours = (9×60) minutes = 540 minutes

So, the ratio in the simplest form

$$\begin{aligned}
 &= 9 \text{ hours} : 36 \text{ minutes} \\
 &= 540 \text{ minutes} : 36 \text{ minutes} \\
 &= 540 : 36 \\
 &= 15 : 1 \text{ (dividing both by 36)}
 \end{aligned}$$

Describe the relationship between ratio and fraction

Ratio is a form of common fraction, where numerator is called antecedent and the denominator is known as consequent.

For example. The fraction $\frac{3}{4}$ is written 3:4 as ratio.

or $\frac{3}{4} = 3 : 4$

Also $\frac{2}{5} = 2 : 5$, $\frac{3}{1} = 3 : 1$, $\frac{25}{39} = 25 : 39$ etc.

Example 1: Two pieces of cloth are of 2 m 50 cm and 75 cm length respectively. Find the ratio between their lengths.

Solution: Let us convert the given lengths in the same units.

As, 1 m = 100 cm,

$$\begin{aligned}
 \text{So, } 2 \text{ m } 50 \text{ cm} &= (2 \times 100)\text{cm} + 50 \text{ cm} \\
 &= (200 + 50)\text{cm} = 250 \text{ cm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence, the ratio of their lengths} &= 2\text{m } 50\text{cm} : 75\text{cm} \\
 &= 250\text{cm} : 75\text{cm}
 \end{aligned}$$

$$= \frac{250}{75} = \frac{10}{3}$$

$$= \frac{10}{3} = 10:3$$

Example 2: Find ratio of the following and write in lowest form.

(i) $\frac{1}{5}$ and $\frac{2}{3}$

(ii) $1\frac{3}{5}$ m and $\frac{7}{10}$ m

(iii) 1.5 and $7\frac{1}{2}$

(iv) Rs. 3.45 and Rs. 6

Solution:

(i) $\frac{1}{5} : \frac{2}{3} = \frac{1}{5} \times 15 : \frac{2}{3} \times 15$

(Multiplying both by LCM of 5 and 3)

$= 3 : 10$

(ii) $1\frac{3}{5} : \frac{7}{10} = \frac{8}{5} : \frac{7}{10}$

$= \frac{8}{5} \times 10 : \frac{7}{10} \times 10$

(Multiplying both by LCM of 5 and 10)

$= 16 : 7$

(iii) $1.5 : 7\frac{1}{2} = 1.5 : \frac{15}{2}$

$= 1.5 \times 10 : \frac{15}{2} \times \frac{2}{1} \quad \text{(Multiplying both by 10)}$

$= 15 : 15 \times 5$

$= 1 : 5 \quad \text{(dividing both by 15)}$

(iv) $3.45 : 6 = 345 : 600$

(As 3.45 is two degree decimal, so we have convert it in to a whole number by multiply 100. Convert both components of ratio into whole numbers so, we multiply both numbers by 100).

$= 69 : 120 \quad \text{(Dividing both by 5)}$

$= 23 : 40 \quad \text{(Dividing both by 3)}$

EXERCISE 6.1

1. Reduce the following into lowest equivalent form.

- (i) $4 : 50$ (ii) $0.8 : 72$ (iii) $3.5 : 4.9$
(iv) $\frac{13}{60} : \frac{7}{15}$ (v) $\frac{5}{6} : \frac{3}{10}$ (vi) $\frac{2}{3} : 5$
(vii) $2\frac{1}{2} : 4$ (viii) $\frac{1}{3} : \frac{1}{6} : \frac{1}{9}$ (ix) $1.5 : 5 : 5.8$
(x) $3\frac{2}{5} : 0.6 : 3.5$ (xi) $\frac{1}{5} : \frac{1}{10} : \frac{1}{15}$ (xii) $\frac{1}{7} : \frac{1}{14} : \frac{1}{21}$

2. Find ratio of the following and write in lowest form.

- (i) Rs 150 and Rs 180 (ii) 250 cm and 1 m
(iii) 700 g and 2 kg (iv) 3 hours and 210 minutes
(v) 5 years and 3 months
(vi) 15 days, 2 weeks and 1 month

3. Convert the following fractions into ratio form.

- (i) $\frac{2}{9}$ (ii) $\frac{5}{6}$ (iii) $\frac{1}{75}$ (iv) 3 (v) $\frac{p}{q}$

4. Represent the following into fractions.

- (i) 1:5 (ii) 2:19 (iii) 8:1 (iv) 75:76 (v) $x : y$

5. In a Science test, 25 students out of 45 students of class VI were passed. Find the ratio between the passed students and total students.
6. Arshad earns Rs 20,000 per month. Find the ratio of his monthly income and expenditure if he saves Rs. 5000 per month.
7. The measures of sides of two squares are 2 cm and 5 cm. Find the ratio of their perimeters.
8. Weight of a sack of flour is 16 kg and the weight of another sack is 14kg 400g. Find the ratio among their weights.
9. Measures of angles of a triangle are 30° , 60° and 90° . Find the ratio of these angles according to the given order.

6.2 PROPORTION

Two equivalent ratios form a proportion.

For example: $3 : 4 = 6 : 8$ is a proportion.

Read as 3 is to 4 equals 6 is to 8.

Know that an equality of two ratios constitutes a proportion, e.g., $a : b :: c : d$, where a, d are known as extremes and b, c are called the means

In a proportion, the equality sign can be replaced by “::”.

So, for the proportion $2:7 = 6:21$, we can write $2:7 :: 6:21$

The two middle elements of a proportion are called **means**, where as the elements at both ends of a proportion are known as **extremes**. For example:

$$\begin{array}{c} \text{Means} \\ \overbrace{2 : 7 :: 6 : 21} \\ \text{Extremes} \end{array}$$

Also, 2, 7, 6 and 21 are called first, second, third and fourth proportionals respectively of the given proportion.

The equality of two given ratios can be proved by the fact that $2:7::6:21$ can also be written as $\frac{2}{7} = \frac{6}{21}$ or $2 \times 21 = 6 \times 7$ or $42 = 42$

This can also be proved with the help of this formula of proportion.

Product of extremes = product of means.

If means are same for example, $1:2 = 2:4$ then 2 is called the mean proportional between 1 and 4.

Also 1, 2 and 4 are said to be in continued proportion. In this case the third element 4 is called a third proportional of the first and second. By using these properties of proportions we can find any missing proportional. As shown in the following examples.

Example 1: Identify means and extremes in the following:

(i) $2 : 3 = 10 : 15$

Solution:

Here

Means are 3 and 10

Extremes are 2 and 15

(ii) $5 : 1 = 10 : 2$

Solution:

Here

Means are 1 and 10

Extremes are 5 and 2

Example 2: Decide whether the following numbers are in proportion or not.

(i) 2, 3, 4, 6

(ii) 5, 10, 15, 20

Solution: (i) 2, 3, 4, 6

Here

$$\begin{array}{c} \overbrace{2 : 3} \\ \underbrace{= 4 : 6} \end{array}$$

Here Product of means = $3 \times 4 = 12$

and product of extremes = $2 \times 6 = 12$

$\therefore$ Product of means = Product of extremes

$\therefore$ 2, 3, 4, 6 are in proportion

Solution: (ii) $5 : 10 = 15 : 20$

Product of means = $10 \times 15 = 150$

Product of extremes = $5 \times 20 = 100$

$\therefore$ Product of means $\neq$ Product of extremes

$\therefore$ 5, 10, 15, 20 are not in proportion

Example 3: Find the value of unknowns in the following proportionals:

(i) $33 : 3 :: x : 27$ (ii) $9 : 6 :: 6 : y$

Solution:

$$(i) \quad 33 : 3 :: x : 27$$

(product of extremes = product of means)

$$\text{or} \quad 33 \times 27 = 3 \times x$$

$$\text{or} \quad \frac{33 \times \cancel{27}^9}{\cancel{3}} = x$$

$$\text{or} \quad x = 33 \times 9$$

$$\text{or} \quad x = 297$$

$$(ii) \quad 9 : 6 :: 6 : y$$

$$\text{or} \quad 9 \times y = 6 \times 6$$

$$\text{or} \quad y = \frac{\cancel{26}^2 \times \cancel{6}^2}{\cancel{9}_3 \times 1}$$

$$\text{or} \quad y = 2 \times 2$$

$$\text{or} \quad y = 4$$

Example 4: Find fourth proportional of 2, 5, 8.**Solution:** Let fourth proportional be x .

So, we have

$$2 : 5 = 8 : x$$

$$\text{or} \quad 2 \times x = 5 \times 8$$

$$\text{or} \quad \frac{2 \times x}{2} = \frac{5 \times 8}{2}$$

$$\text{or} \quad x = 20$$

So, fourth proportional is 20.

Example 5: Find the mean proportional of 4 and 9.**Solution:** Let x be the mean proportional, then

$$4 : x :: x : 9$$

$$\text{or} \quad 4 \times 9 = x^2$$

$$\text{or} \quad x^2 = 36$$

$$\text{or} \quad x^2 = 6^2$$

$$\text{or} \quad x = 6$$

Hence 6 is the mean proportional of 4 and 9.

EXERCISE 6.2

- Identify means and extremes in the following:
 (i) $2 : 5 = 8 : 20$ (ii) $3 : 4 = 6 : 8$ (iii) $a : b = c : d$
- Decide whether the four numbers given in each of the following are in proportion or not:
 (i) 18, 24, 30 and 40 (ii) 14, 19, 3 and 4
 (iii) 6, 8, 12 and 16 (iv) 15, 20, 16 and 21
 (v) 20, 30, 40 and 50 (vi) 21, 57, 28 and 76
- Find the value of x if the proportion $2 : x = 3 : 7$.
- Find the fourth proportional in the following:
 (i) 2, 3 and 6 (ii) $\frac{11}{24}$, $\frac{7}{15}$ and $\frac{5}{8}$ (iii) 16, 12 and 8
 (iv) 76, 28 and 57 (v) 36, 45 and 4 (vi) 40, 30 and 24
- Find the mean proportion in the following:
 (i) 15 and 60 (ii) 18 and 32 (iii) 28 and 63
 (iv) 27 and 12 (v) 40 and 90 (vi) 44 and 99

Find proportion (direct and inverse)

(1) Direct proportion:

Consider the following chart.

Quantity of Pencils	Cost
4	Rs 20
3	Rs 15
2	Rs 10
1	Rs 5

What is the proportion of number of pencils and their cost.

In this chart, we can easily observe that the cost of pencils increases or decreases with the corresponding increase or decrease in the quantity of pencils.

Hence we can say that when two quantities are related in such a way that if one quantity increases in a given ratio, the other also increases in the same ratio (and vice versa).

In this situation, there is a **Direct Proportion** between both quantities.

Note: In the above chart, the ratio between 1 pencil and 2 Pencils is same as between the cost Rs 5 and Rs 10.

Examples:

- (i) More construction, more materials
- (ii) Less money, less shopping
- (iii) More students, more teachers
- (iv) Less workers, less work

(2) Inverse Proportion:

The following chart is showing number of persons required to complete a work in the given time.

No. of Persons	Time
2	40 hours
4	20 hours
8	10 hours
16	5 hours

What type of proportion is between number of person and time required.

In the above table, we observe that by increasing the number of persons the time required is decreased. (and vice versa).

Hence we can say that when two quantities are related in such a way that if one quantity increases in a given ratio, the other decreases in the same ratio.

In this situation, there is an **inverse proportion** between both quantities.

Note: In the above chart the ratio between 2 persons and 4 persons is same as between 20 and 40.

Examples:

- (i) Less speed; more time taken
- (ii) More speed; less time taken
- (iii) More time; less workers required
- (iv) Less time; more workers required

Solve real life problems involving direct and inverse proportion

Consider the following examples.

Example 1: If 20 metre cloth is used to make 4 dresses. How much cloth is required to make such 15 dresses?

Solution: Suppose the required cloth is x metre

Dresses

$$\begin{array}{c} 4 \\ \downarrow \\ 15 \end{array}$$

(More cloth more dresses)

Cloth in metres

$$\begin{array}{c} 20 \\ \downarrow \\ x \end{array}$$

Direct Proportion

$$4 : 15 :: 20 : x$$

As product of means = Product of extremes

$$\text{or } 4 \times x = 15 \times 20$$

$$\text{or } x = \frac{15 \times 20}{4}$$

$$\text{or } x = 75$$

Hence the required cloth is 75 metre.

Teacher's Note

Teacher should discuss other examples from daily life with the students and help them in solution.

Example 2: If 3 pipes can fill a tank in 80 minutes. How long will it take to fill the tank if 5 pipes are used?

Solution:

Let x minutes are required.

Pipes	Time in minutes
↑ 3	↑ 80
5	x

(More time less pipes)

Inverse Proportion

$$\overbrace{5 : 3 :: 80 : x}$$

As product of means = Product of extremes

$$\text{or } x \times 5 = 3 \times 80$$

$$\text{or } x = \frac{3 \times \overset{16}{\cancel{80}}}{\cancel{5}_1}$$

$$\text{or } x = 48$$

Hence 48 minutes are required.

EXERCISE 6.3

- Ahmed secured 60 marks in a paper of 100 marks. What shall be his score if the paper were of 75 marks?
- The weight $\frac{5}{9}$ of a piece of metal is 35 kg. What is the weight $\frac{2}{7}$ of the piece?
- On a map, 80 km are represented by 5 cm. If the distance between two cities on the map is 15 cm. Find the actual distance between them.

4. If a dozen eggs costs Rs 60, find the cost of 32 eggs.
5. Tabassum earns Rs. 42,000 in a month and spends Rs 39,800. Find the ratio of her savings to her income. Also find the ratio of her expenditure to her income.
6. If Naveed can read 45 pages of a book in 75 minutes, find the time he takes to read the book of 876 pages.
7. A book covers 130 pages with 24 lines on every page. If we make the same book of 90 pages with increasing number of lines. How many lines would be there on each page?
8. If 80 workers complete a work in 6 days. Find the number of days required to complete the same by 10 workers.

REVIEW EXERCISE 6

1. **Find the ratio in lowest form between the following:**
 - (i) Length and breadth of your copy.
 - (ii) Number of students present and absent in your class.
 - (iii) Weight in kg of your two class mates.
 - (iv) Daily pocket money of your three friends.
 - (v) Number of pages of the text books of your class of Mathematics, English, and Social Studies.
2. **Write equivalent ratio for:**
 - (i) $2 : 3 =$ _____
 - (ii) $10 : 8 =$ _____
 - (iii) $1 : 4 =$ _____
 - (iv) $21 : 35 =$ _____

3. Find the value of x in each of the following:

- (1) $2 : 5 :: x : 10$ (2) $5 : x :: 10 : 14$
(3) $15 : 30 :: 1 : x$ (4) $x : 12 :: 3 : 4$
(5) $x : 200 :: 18 : 25$ (6) $132 : 216 :: 33 : x$

4. Which of the following are inverse proportions?

- (i) The number of copies you buy and their total cost.
(ii) The number of men doing a job and the time taken to finish it.
(iii) The amount of time taken in a journey and speed of the vehicle.
(iv) The number of boxes and the number of Pencils packed in them.
- 5.** Out of 670 students in a school, 350 are boys and rest are girls. Find the ratio of:
(i) Boys to total students. (ii) Girls to total students.
(iii) Girls to boys of the school. (iv) Boys to girls of the school.
- 6.** Rehana can write 26 words in 2 minutes. How much time she will take to write 325 words.
- 7.** 12 kg milk costs Rs 1200. What is the cost of $13 \frac{1}{2}$ kg milk?
- 8.** 24 metre high wall is made by 16 workers in a particular time. How many workers will construct a 18 metre high wall in the same time?
- 9.** Food was stored for 600 persons for 18 days. How many persons will eat the same food in 27 days?

SUMMARY

- Comparison between magnitudes of two quantities of the same kind is called ratio.
- Ratio of two numbers a and b , where $b \neq 0$, is written as

Antecedent $\frac{a}{b}$ or $\frac{a}{b}$ Numerator
 Consequent $\frac{a}{b}$ Denominator

- Order of ratio is important i.e. $a : b \neq b : a$ in general
- A ratio remains unchanged if all its elements are multiplied (or divided) by the same number.
- Equality of two ratios is called proportion.
- If a, b, c, d are in proportion, then,

$$a : b :: c : d \text{ or } a : b = c : d$$

Where a, b, c and d are called the First, Second, Third and Fourth proportionals respectively.

- In a, b and c are in continued proportion then,

$$a : b :: b : c$$
- For every $a : b :: c : d$, $ad = bc$
 Here a and d are extremes and b and c are means we have,
 Product of extremes = Product of means
- When two related quantities, increase or decrease together in the same ratio, they are called directly proportional to each other.
- When in two related quantities, increase in one causes decrease in the other, and decrease in one causes increase in the other, then they called inversely proportional to each other.